Unpacking U.S. Traffic Mortality Disparities

DATA 608 — Story 3: Geography, EMS Capacity, Policy, and Infrastructure

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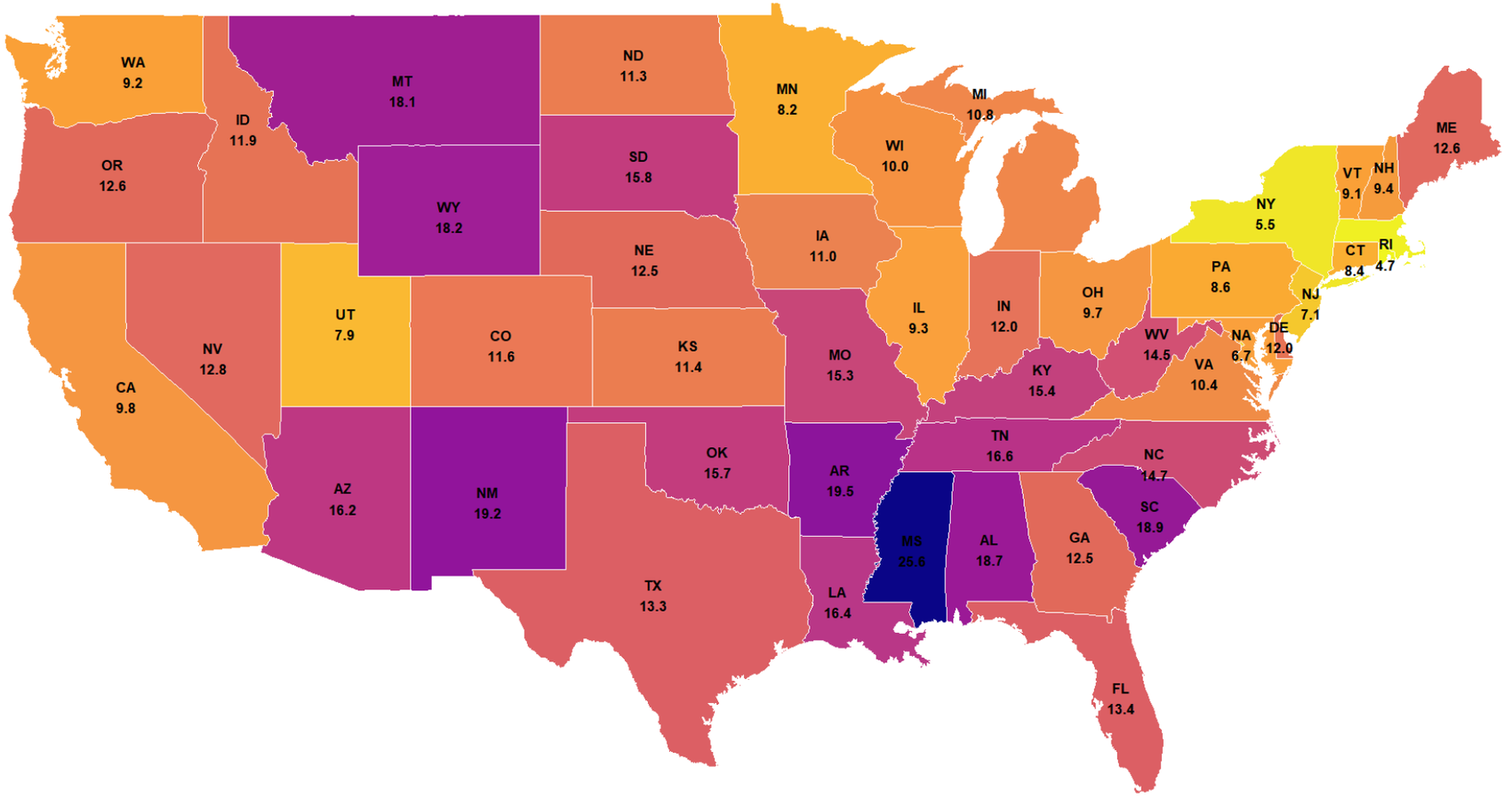
2026-10-08

Introduction & Contextual Background

- Welcome to an in-depth exploration of traffic mortality disparities across the United States. This presentation examines the underlying geographic, structural, and behavioral factors that contribute to fatal motor vehicle crashes.
- While overall transportation trends often look at national aggregates, breaking down the data reveals stark disparities across states, urban versus rural divides, and vulnerable road user populations.

U.S. Road Mortality Rate by State (2024)

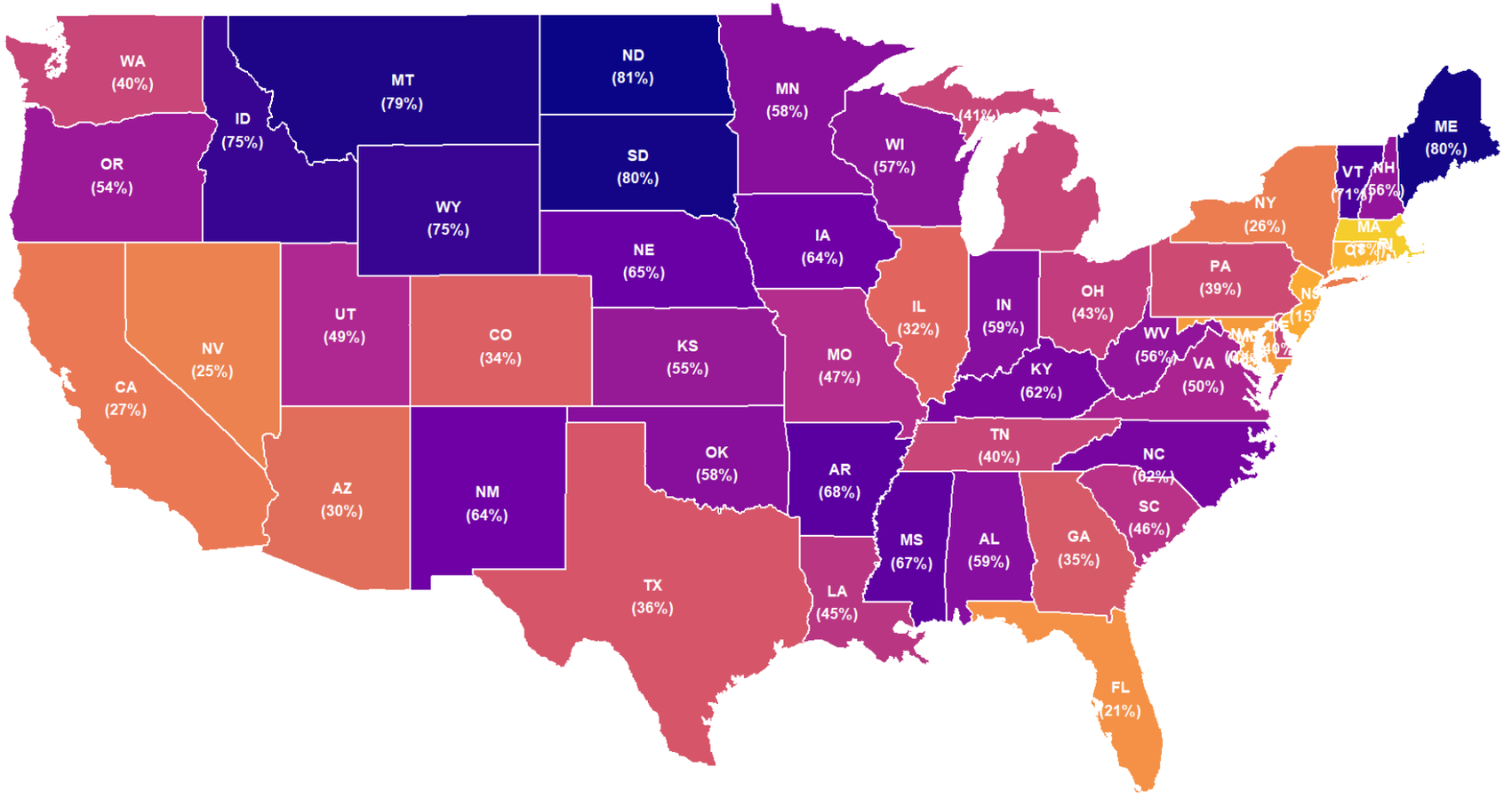
Population-adjusted motor vehicle crash fatalities per 100,000 residents across the contiguous U.S.



Source: FARS 2024 State Summary Data

Share of Motor Vehicle Fatalities on Rural Roads by State (2024)

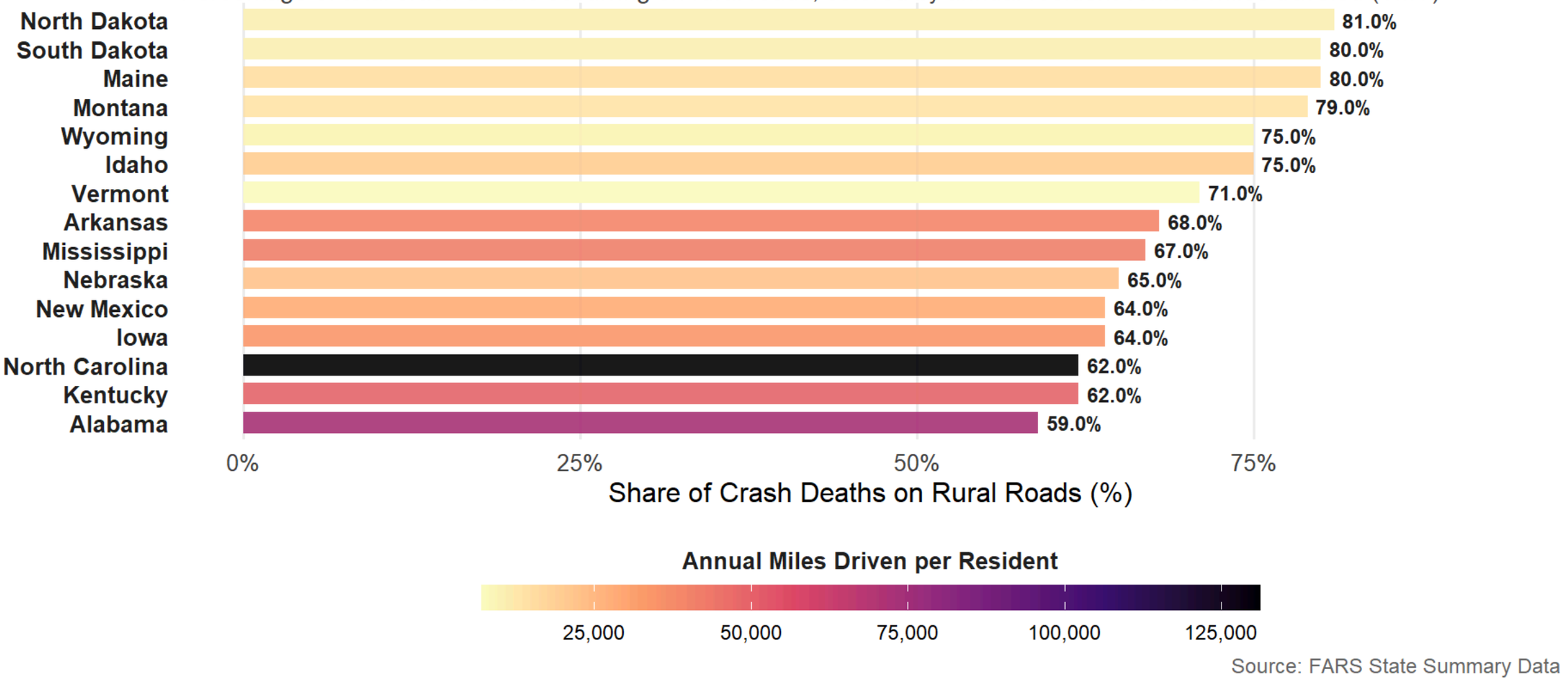
Geographic distribution showing proportion of total traffic deaths on rural roadways



Source: FARS 2024 State Summary Data

Top 15 U.S. States by Rural Traffic Fatality Share (2024)

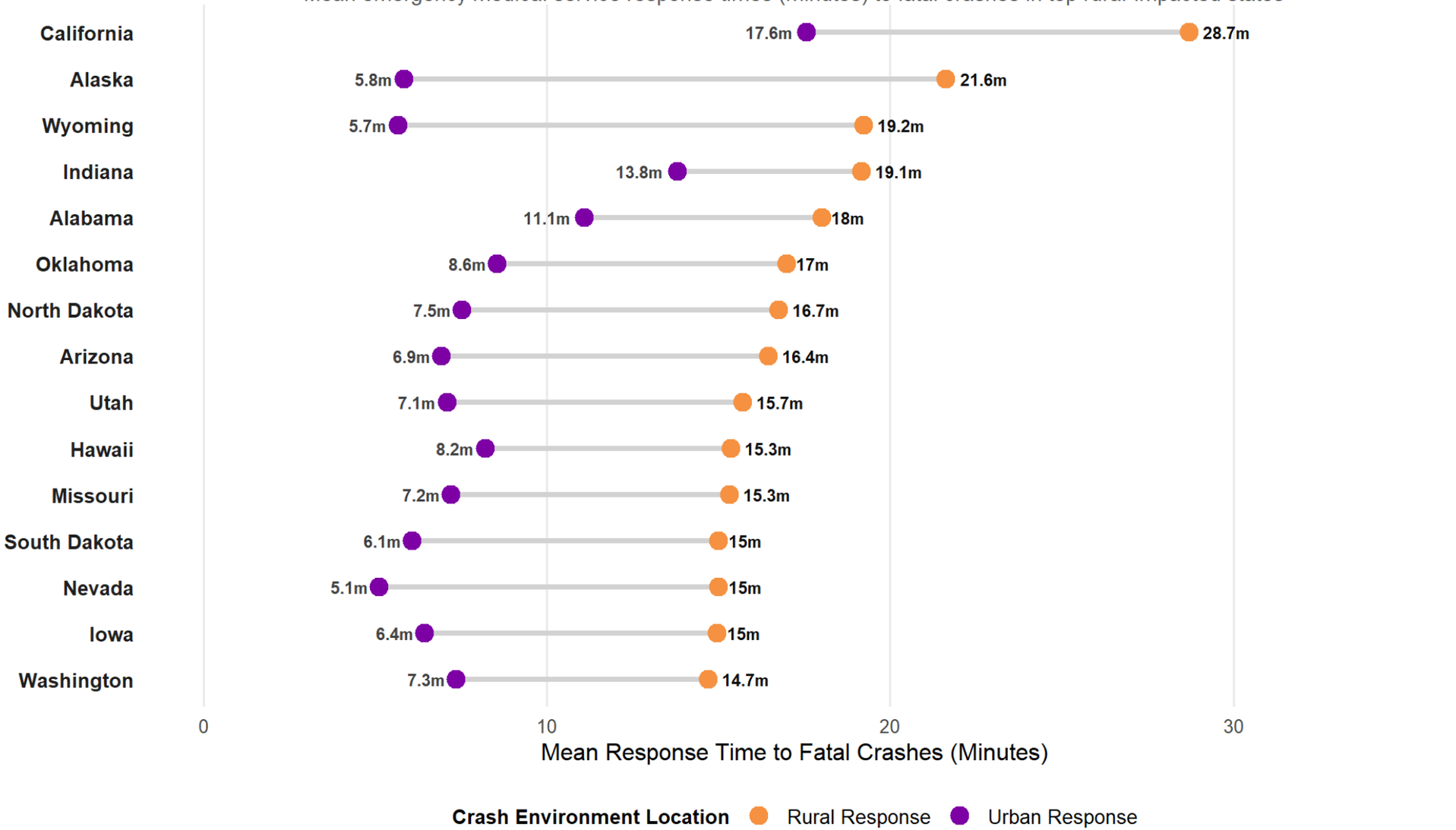
Percentage of total crash deaths occurring on rural roads, colored by annual resident vehicle miles traveled (VMT)



Source: FARS State Summary Data

The 'Golden Hour' Gap: Rural vs. Urban EMS Response Times (2024)

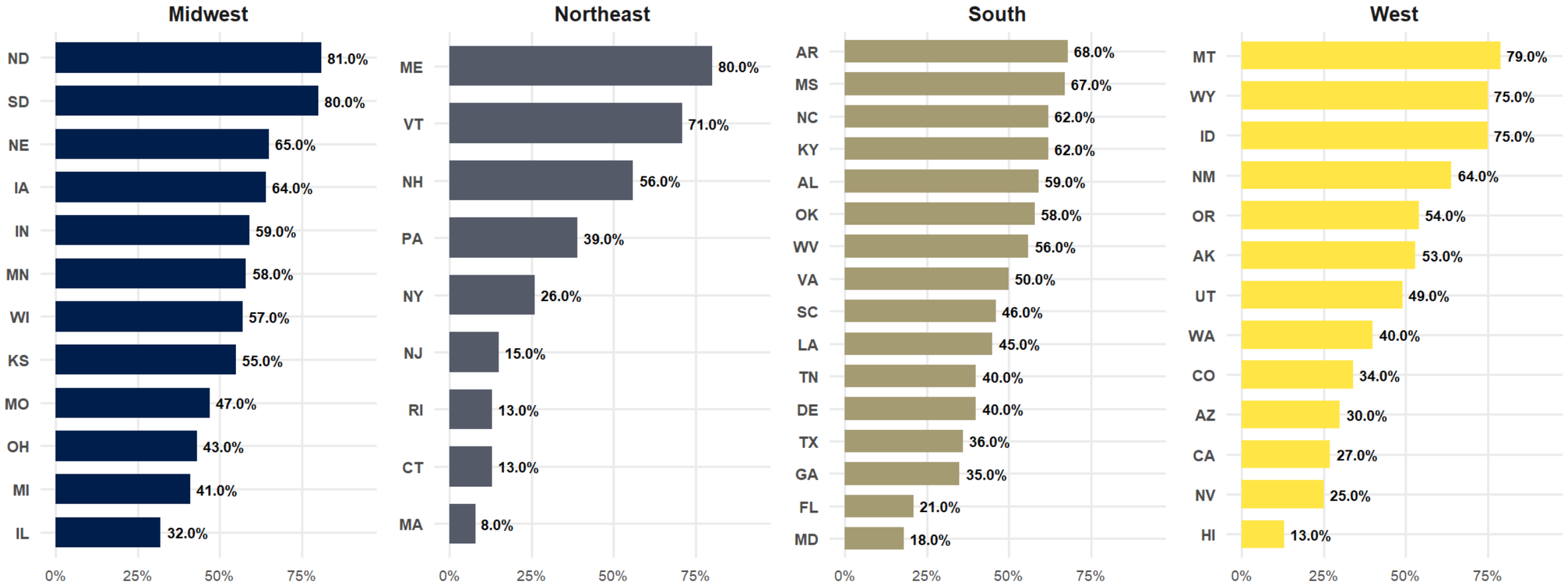
Mean emergency medical service response times (minutes) to fatal crashes in top rural-impacted states



Source: FARS 2024 State Summary Data

Dark Highways & Fatal Accident Disparities by Region (2024)

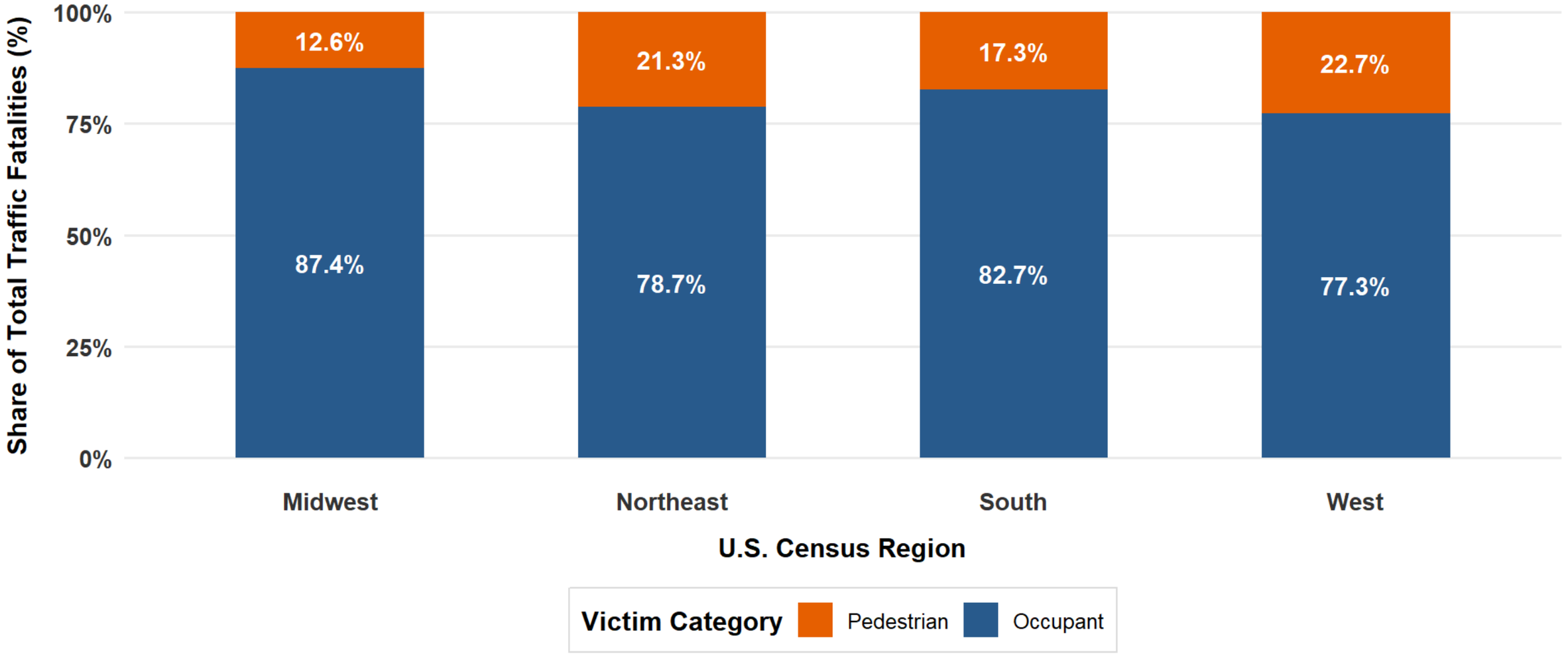
Unlit rural corridors in Southern and Western states show the highest fatal crash shares



Source: FARS State Summary Data

Regional Differences in Vulnerable Road User Fatalities (2024)

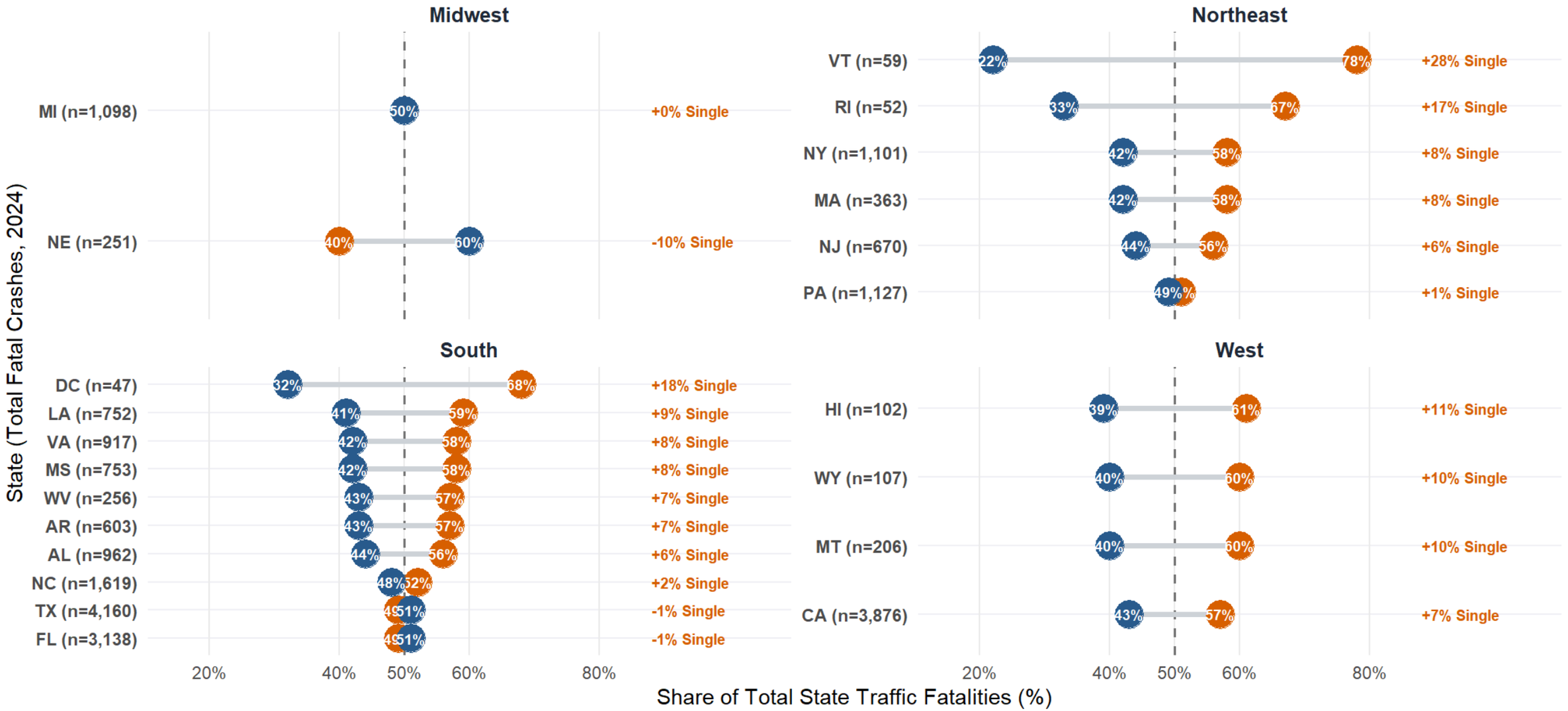
West (22.7%) and Northeast (21.3%) exhibit significantly higher pedestrian death shares



Source: FARS / IIHS State Master Dataset 2024

Crash Environment Disparities: Single vs. Multiple Vehicle Fatalities (IIHS 2024)

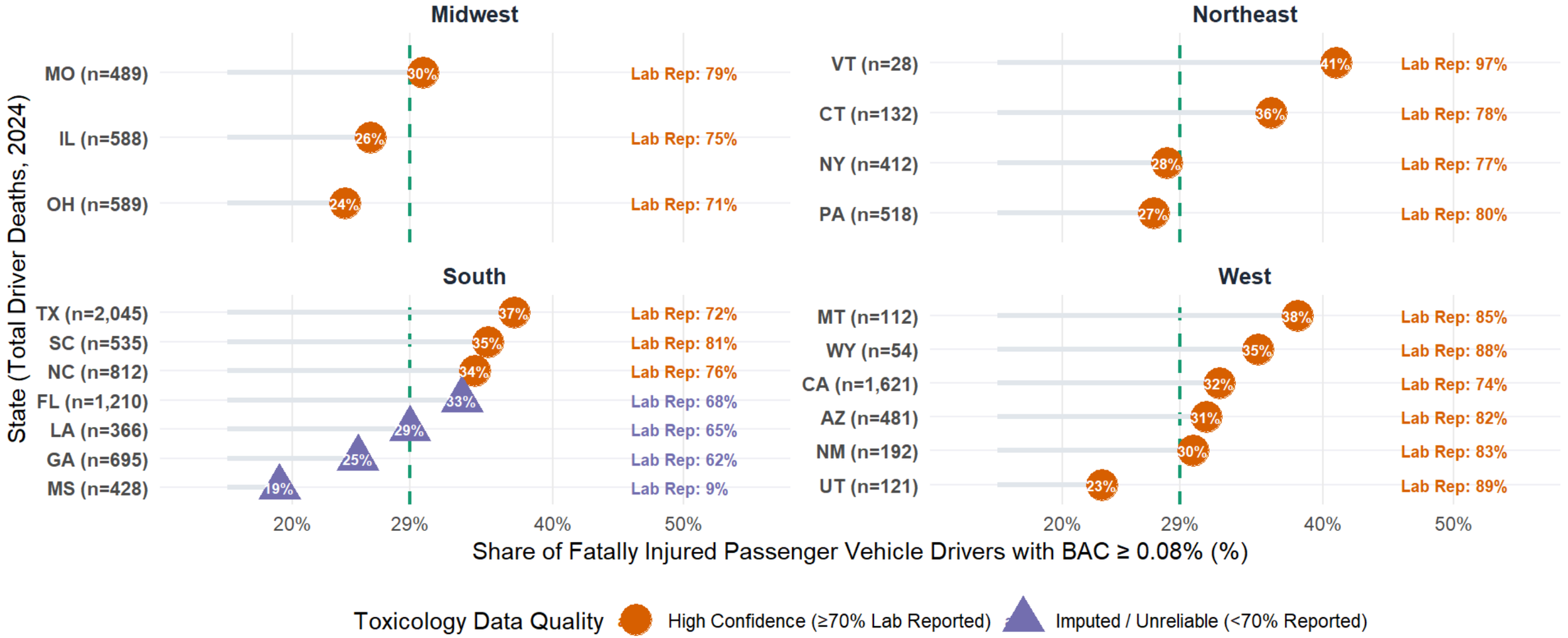
Orange circles (Single-Vehicle) indicate rural/dark highway risks; Blue circles indicate multi-vehicle collisions



Source: IIHS Fatality Facts 2024 | U.S. DOT Fatality Analysis Reporting System (FARS)

Alcohol Impairment in Driver Fatalities by State (IIHS 2024)

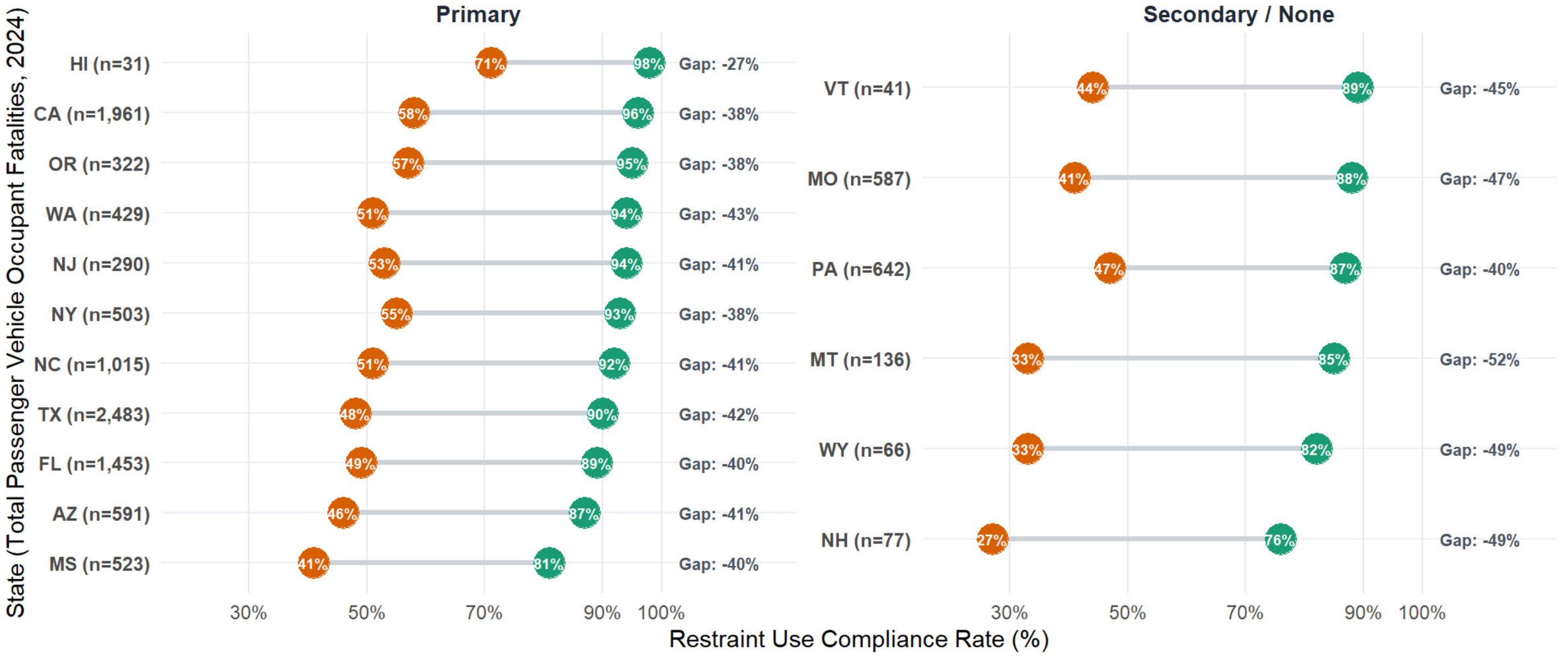
Vermont (41%) and Montana (38%) lead alcohol-impaired driver deaths; Green dashed line indicates US average (29%)



Source: IIHS Fatality Facts 2024 | U.S. DOT Fatality Analysis Reporting System (FARS)

The Seat Belt Compliance Gap: Observed Use vs. Fatality Restraint Rates (IIHS 2024)

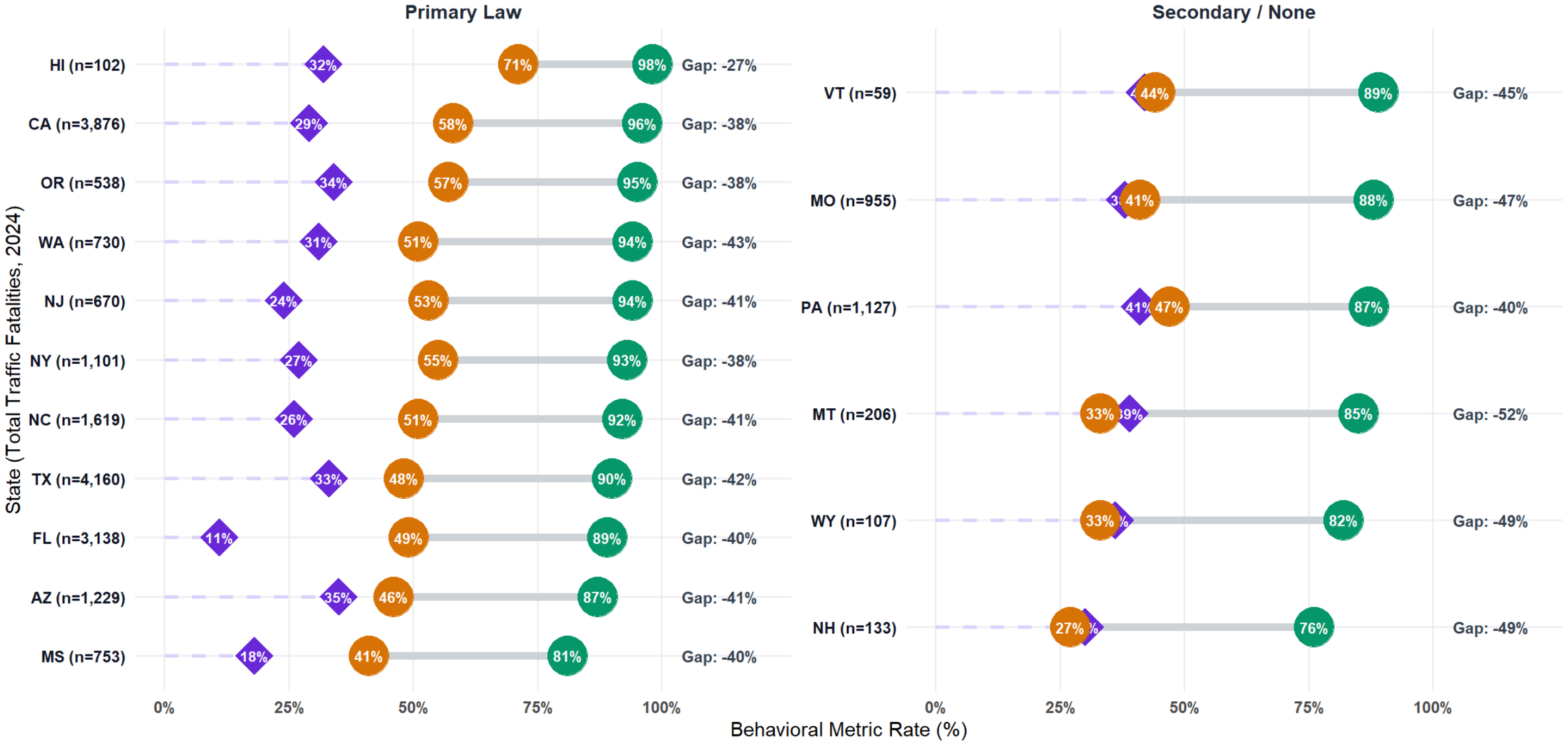
Green circles = Observed Daytime Belt Use; Orange circles = Restrained % among Fatal Occupants. Primary law states maintain higher fatal restraint



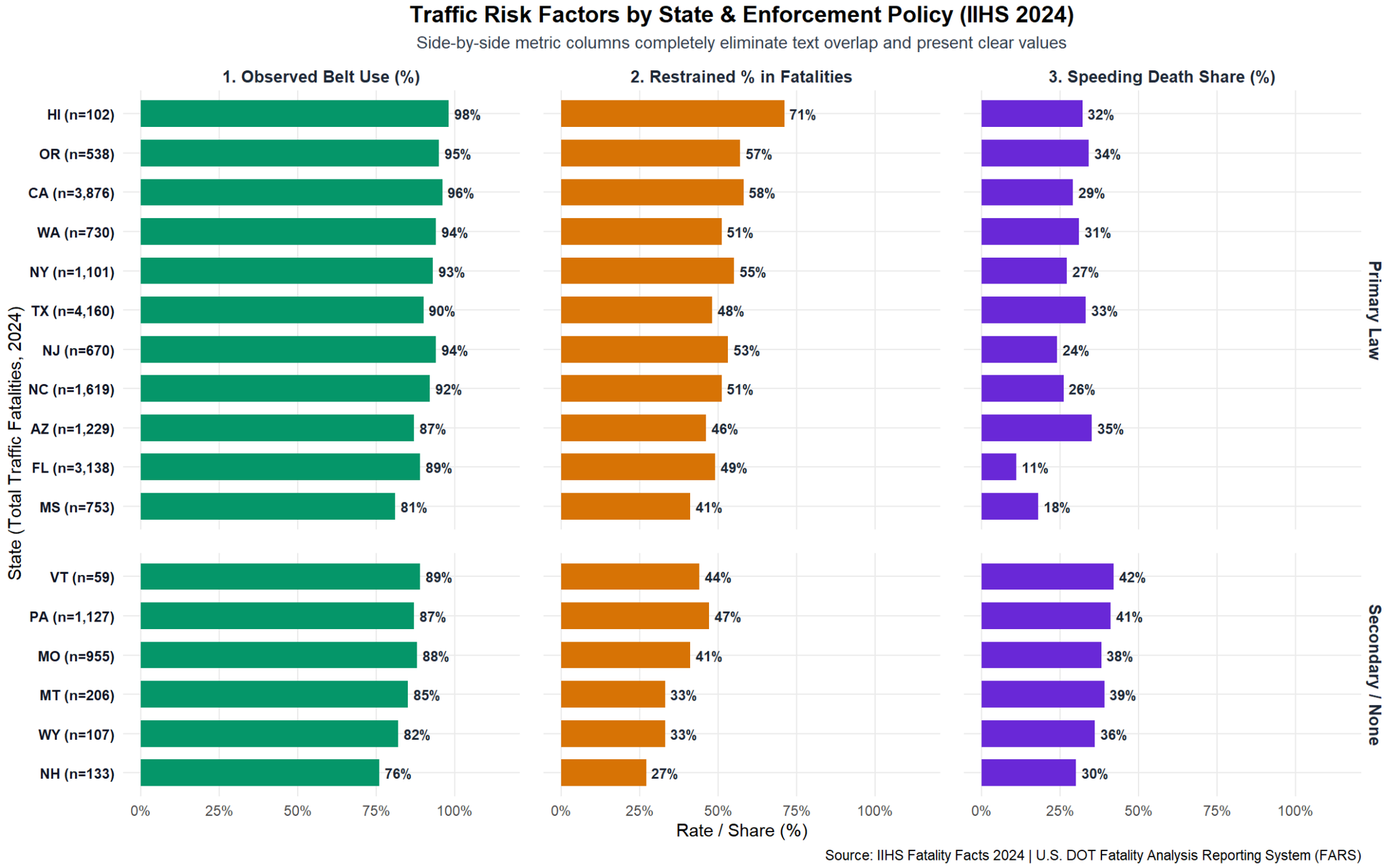
Source: IIHS Fatality Facts 2024 | U.S. DOT Fatality Analysis Reporting System (FARS)

Behavioral Risk Factors: Seat Belt Gap & Speeding Involvement (IIHS 2024)

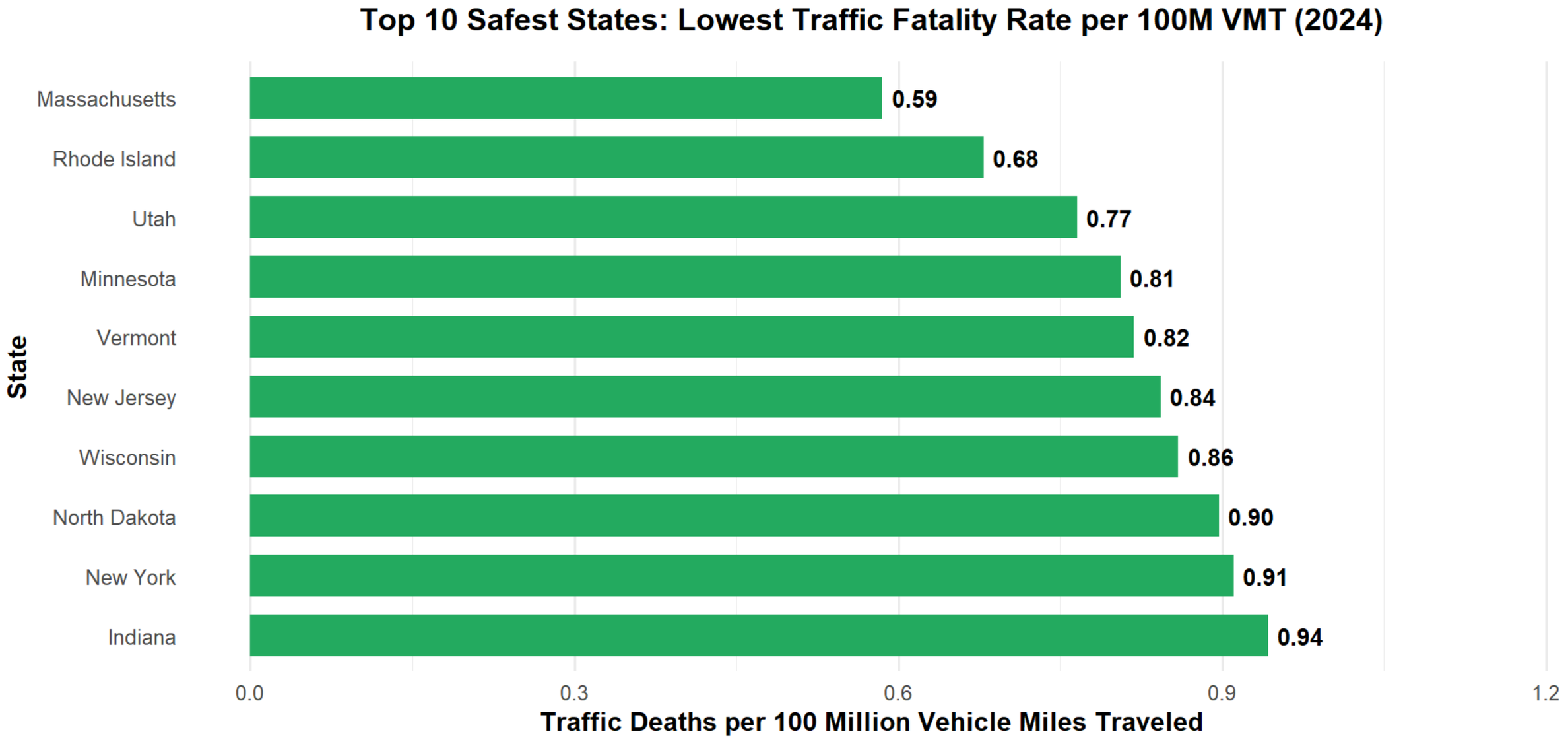
Green Circle = Observed Belt Use | Orange Circle = Restrained % in Fatalities | Purple Diamond = Speeding Share



Source: IIHS Fatality Facts 2024 | U.S. DOT Fatality Analysis Reporting System (FARS)

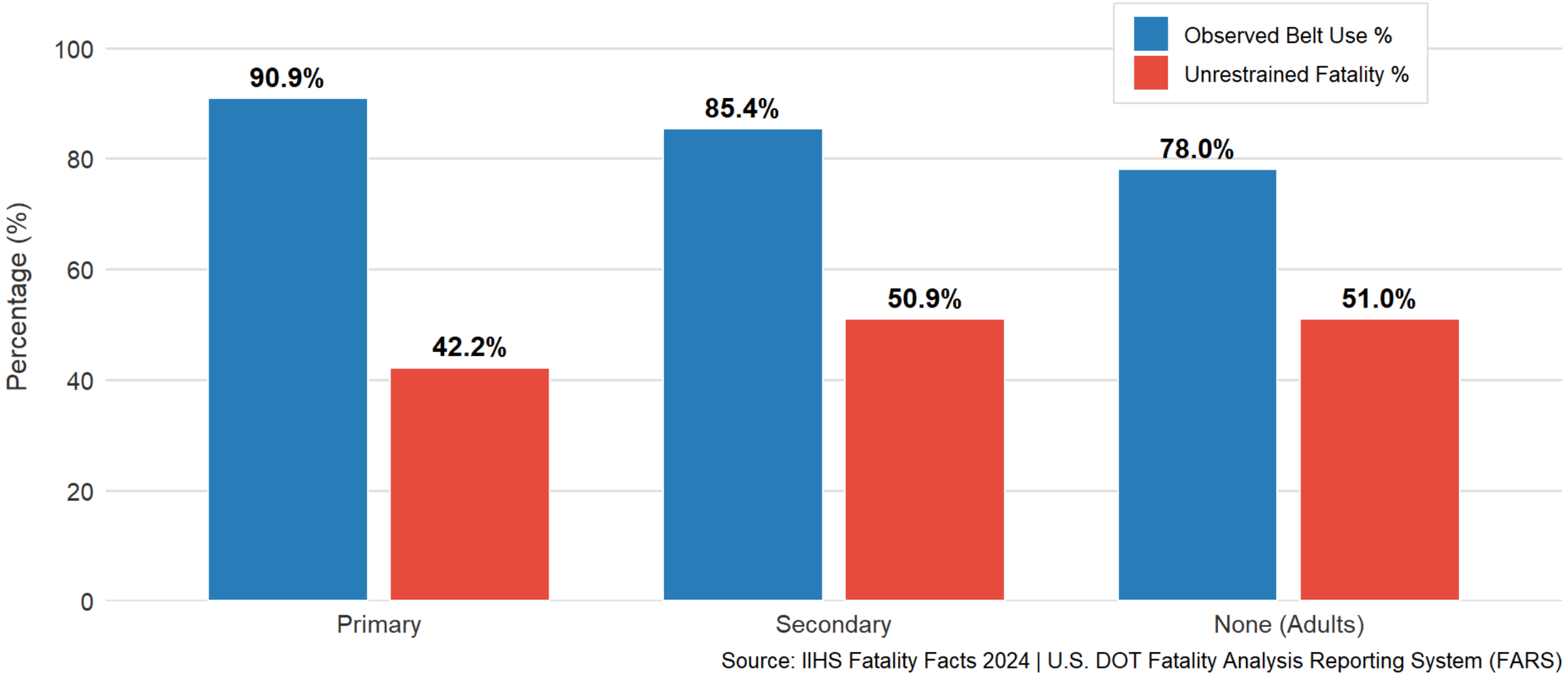


Source: IIHS Fatality Facts 2024 | U.S. DOT Fatality Analysis Reporting System (FARS)



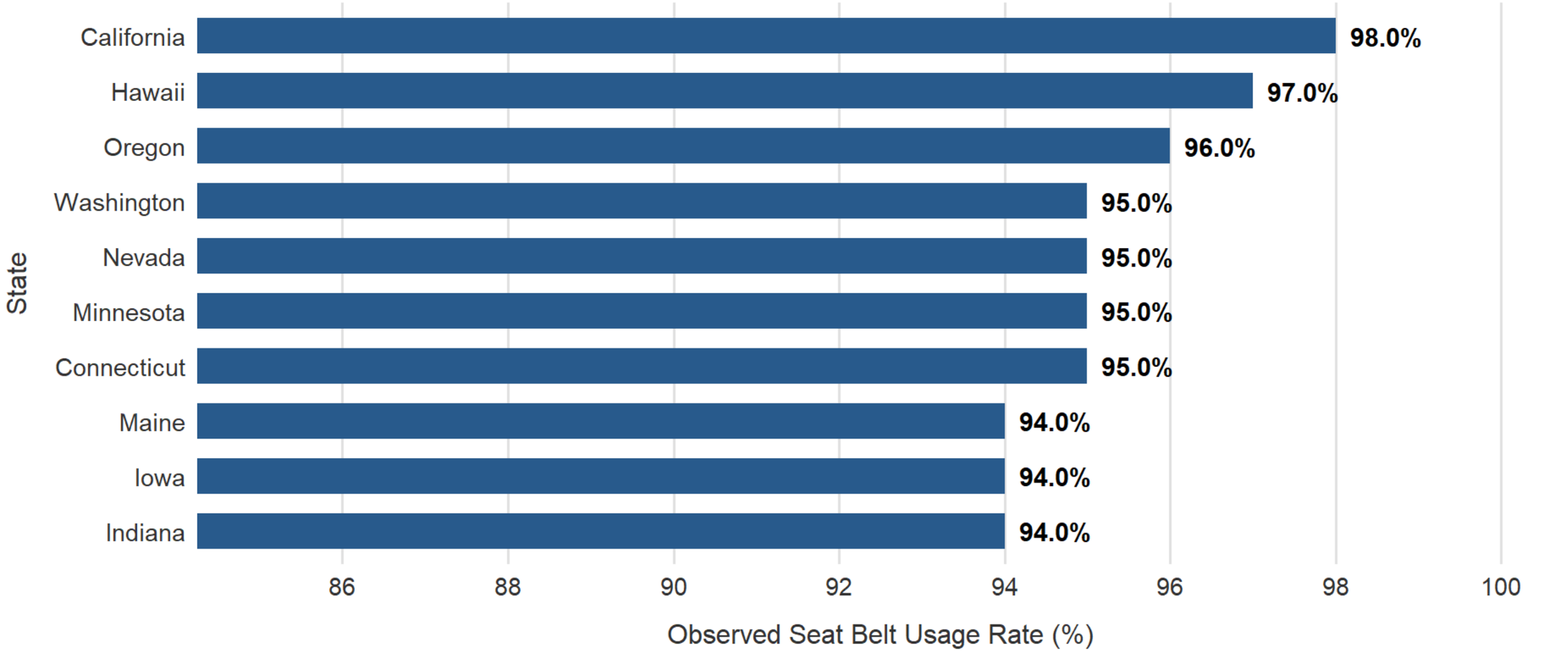
Source: IIHS Fatality Facts 2024 | U.S. DOT Fatality Analysis Reporting System (FARS)

Impact of Seat Belt Enforcement Laws on Usage & Fatality Rates (2024)



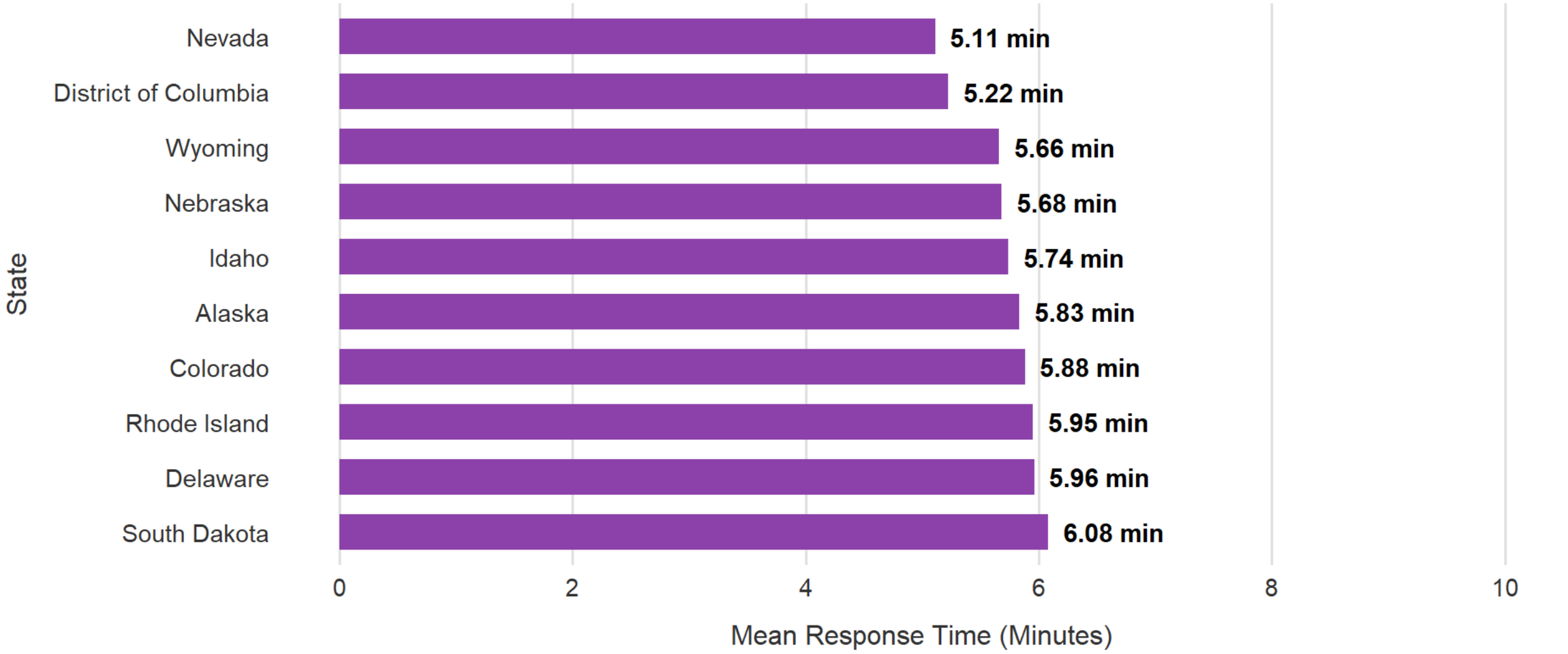
Source: IIHS Fatality Facts 2024 | U.S. DOT Fatality Analysis Reporting System (FARS)

Top 10 States with Highest Observed Seat Belt Compliance (%) (2024)



Source: IIHS Fatality Facts 2024 | U.S. DOT Fatality Analysis Reporting System (FARS)

Top 10 States with Fastest Urban EMS Response Times (2024)



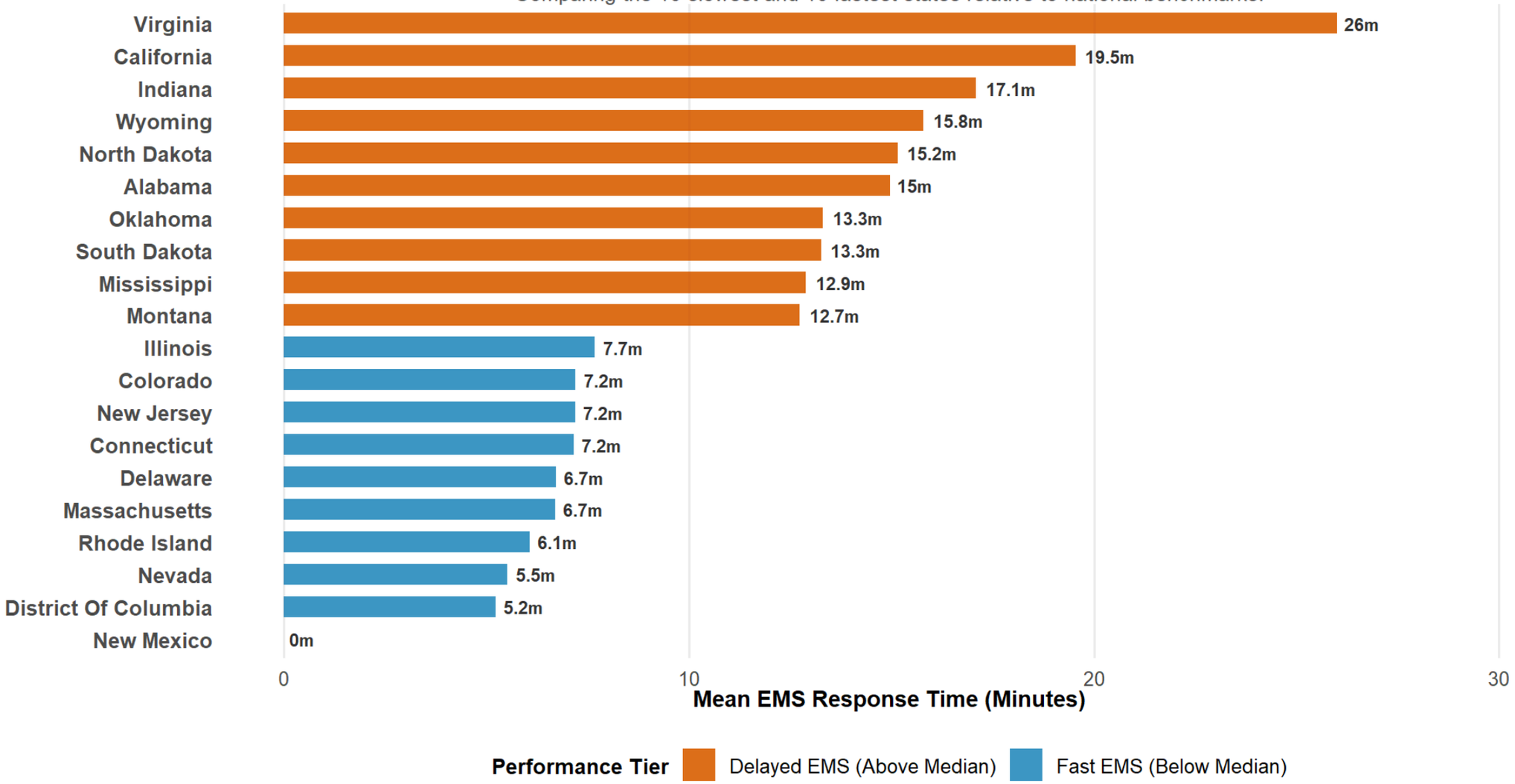
Source: IIHS Fatality Facts 2024 | U.S. DOT Fatality Analysis Reporting System (FARS)

Key Insights & Disparities

- The data highlights a persistent gap in safety outcomes between different regions, often correlated with infrastructure design, emergency response times, and state-level policy enforcement.
- Special attention is given to the disproportionate impact on non-motorized road users, emphasizing the need for equity-centered infrastructure design (such as Vision Zero initiatives).

Healthcare Pillar: Extreme State Variations in EMS Response Times (2024)

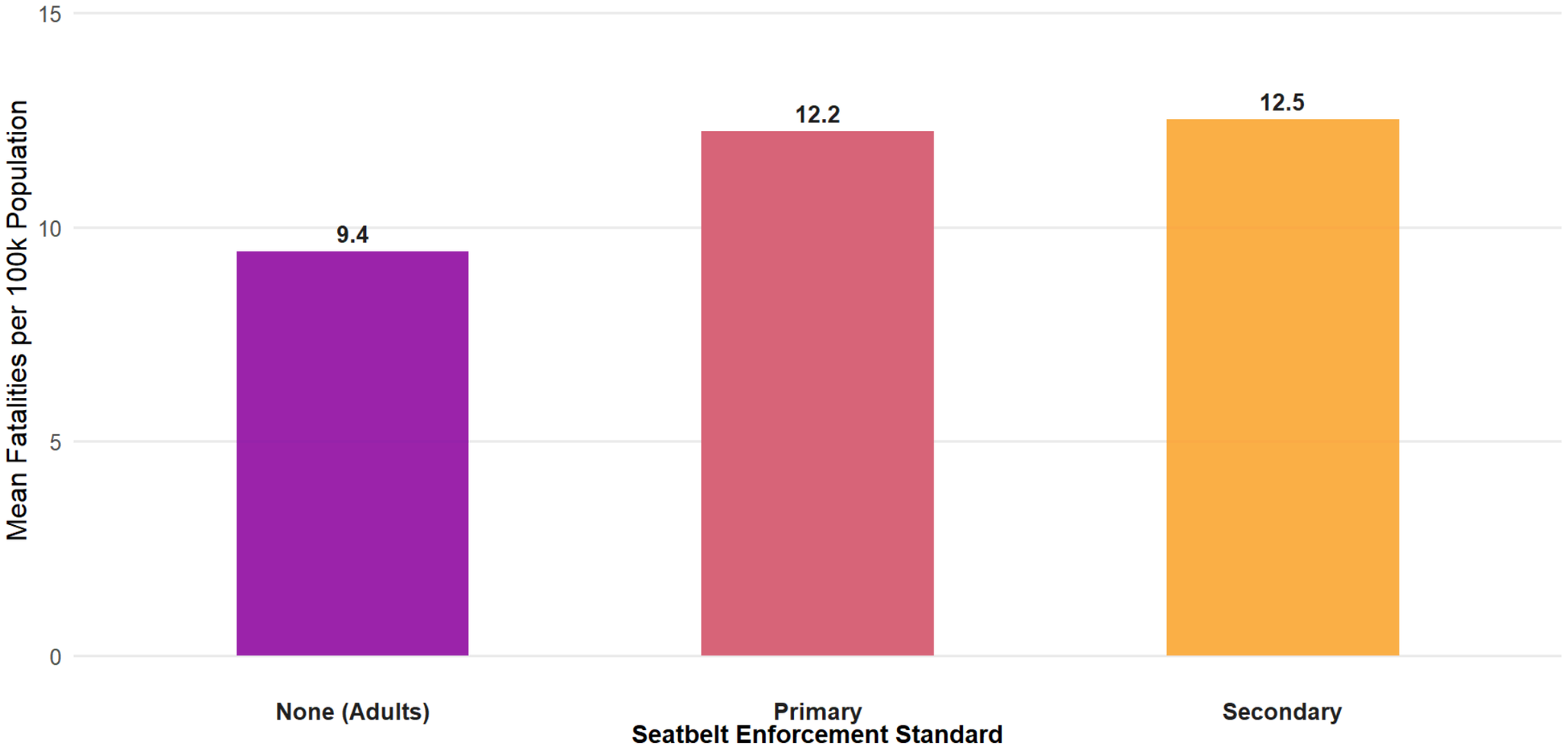
Comparing the 10 slowest and 10 fastest states relative to national benchmarks.



Source: FARS State Summary Data

Policy Pillar: Impact of Seatbelt Enforcement Standards (2024)

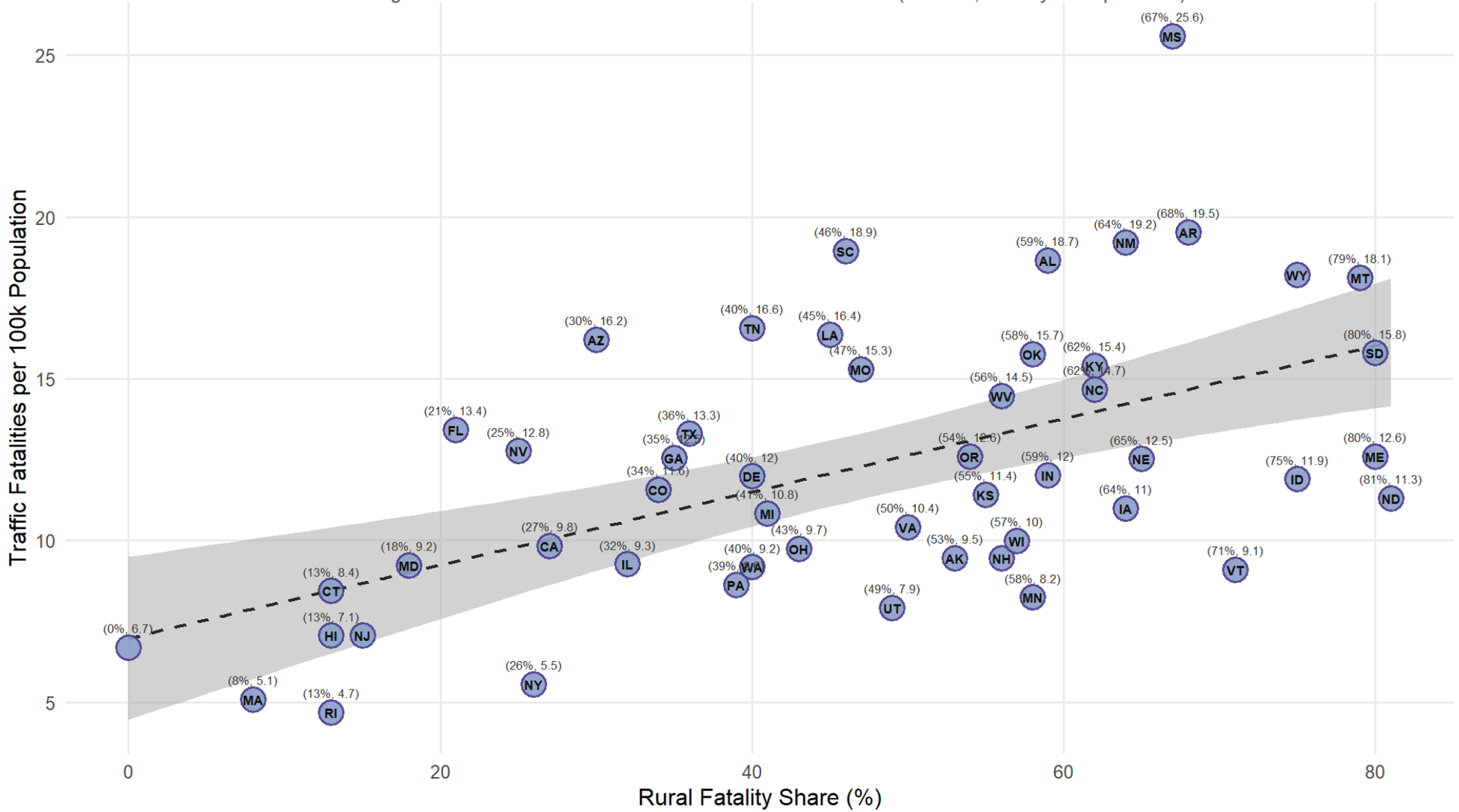
Comparing average state traffic fatality rates across all legislative enforcement categories.



Source: FARS State Summary Data

Infrastructure Pillar: Rural Road Share vs. Crash Mortality (2024)

Showing state abbreviation inside markers and exact data values (Rural %, Fatality Rate per 100k).



Source: FARS State Summary Data

Conclusion & Policy Recommendations

- Mitigating these disparities requires moving beyond traditional enforcement toward systemic changes, prioritizing safer road geometries, lower speed limits in densely populated areas, and data-driven public safety campaigns.
- By understanding where and why these fatalities are concentrated, policymakers and urban planners can better target interventions to save lives.